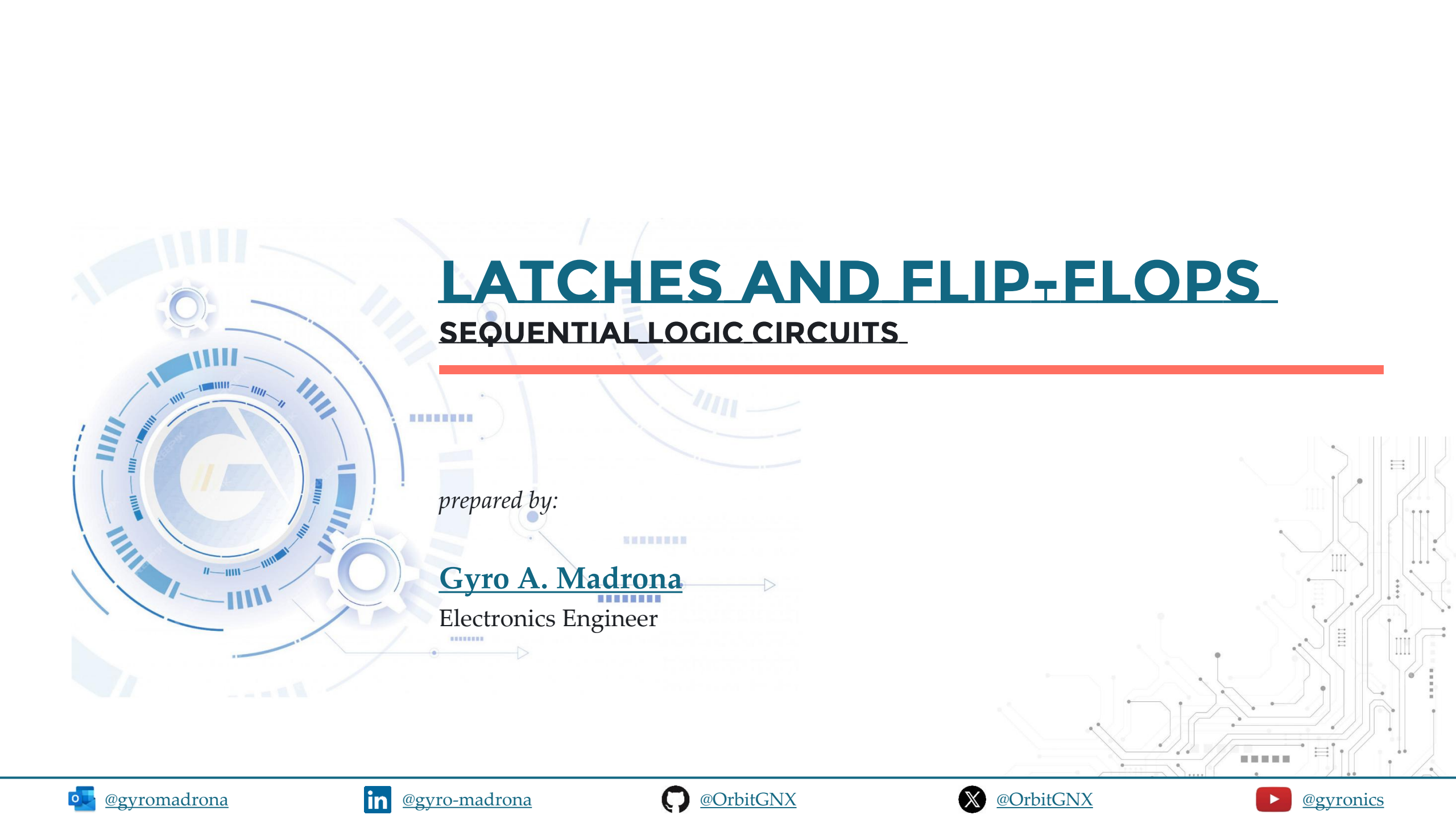




LATCHES AND FLIP-FLOPS

SEQUENTIAL LOGIC CIRCUITS

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

D Latch/Flip-Flop

SR Latch

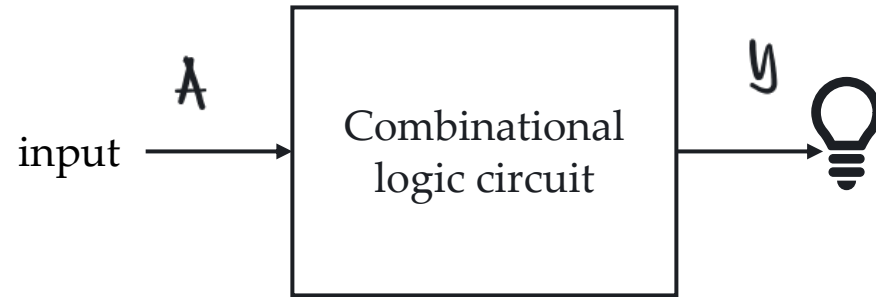
JK Flip-Flop



D LATCH/FLIP-FLOP

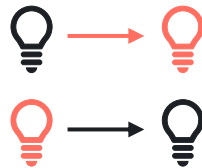


SOMETHING WE CAN'T BUILD (YET)



When the input = HIGH

- if the light is OFF, it turns ON
- if the light is ON, it turns OFF



A	y
0	0
1	1
1	0

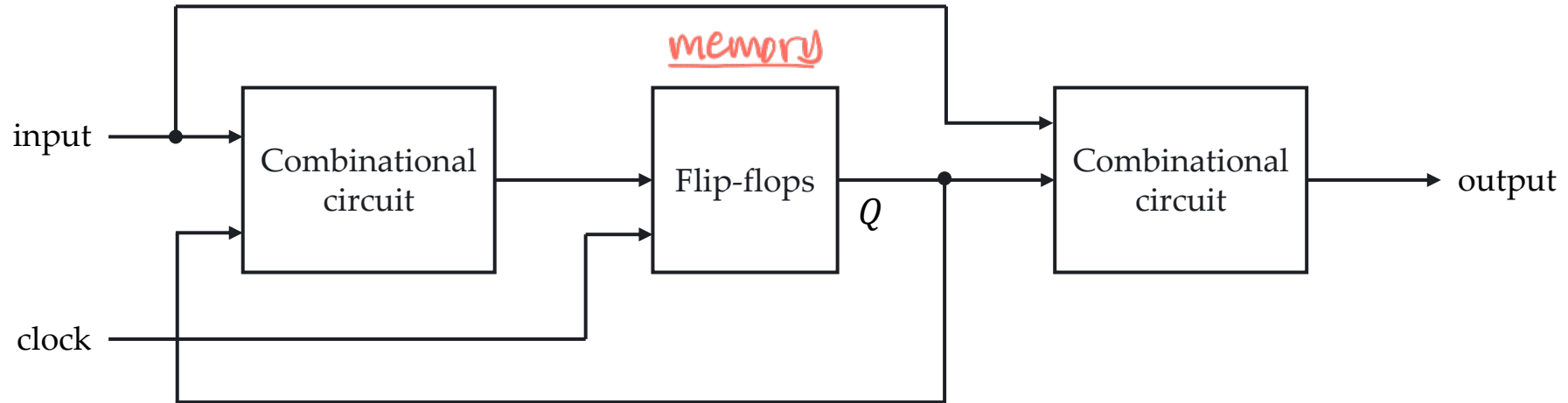
Minterm
 $y = A$
Wrong

The output of combinational logic circuit depends only on the present inputs. It **doesn't remember** whether the light was ON or OFF before.



SEQUENTIAL CIRCUITS

Sequential circuits are digital logic circuits whose outputs depend on the past behavior of the circuit, as well as on the present values of inputs.

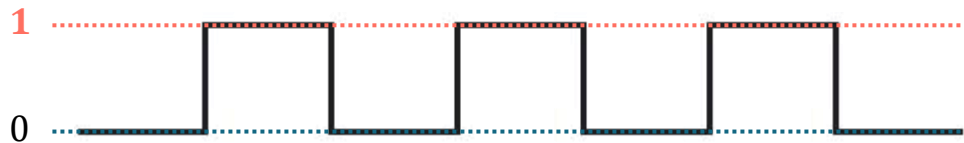
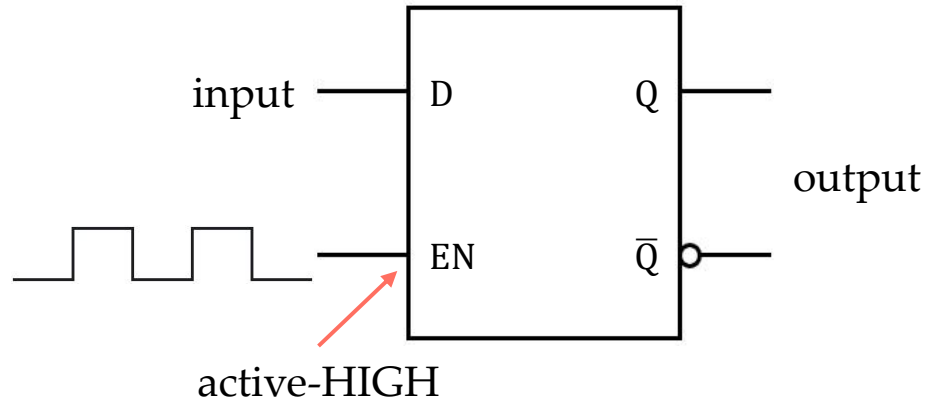


General form of a sequential circuit

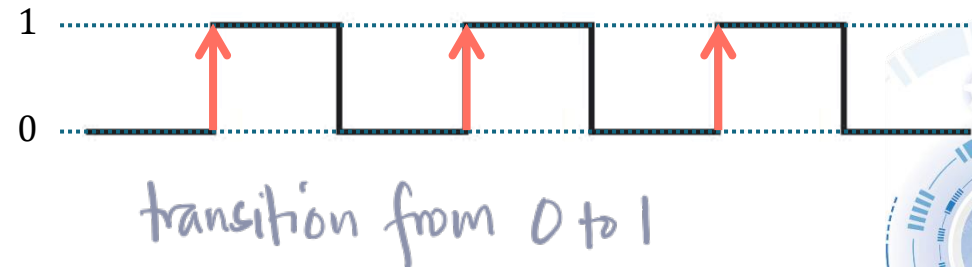
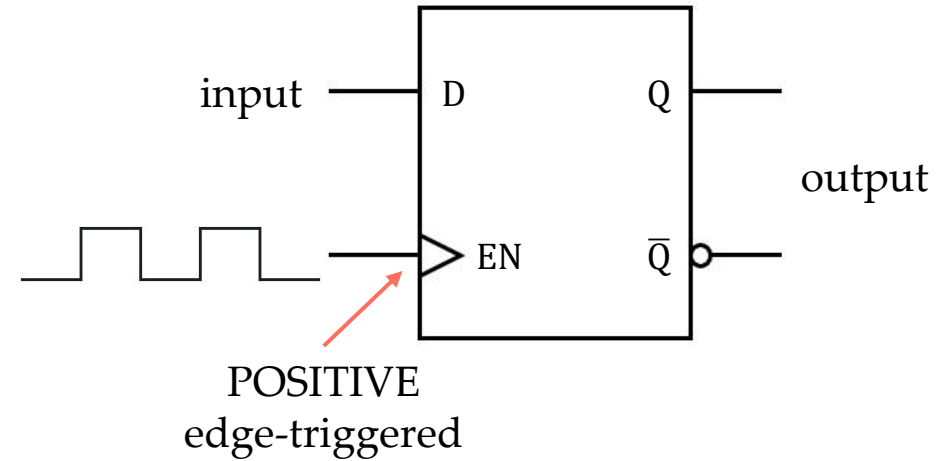


LATCH AND FLIP-FLOP

Latch is a level sensitive (HIGH or LOW) device.

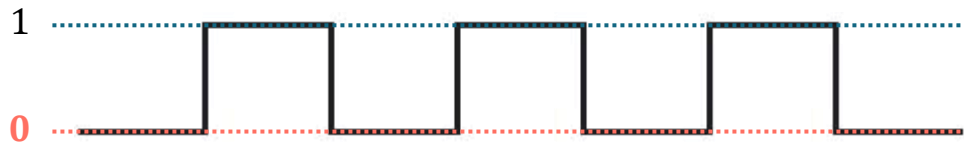
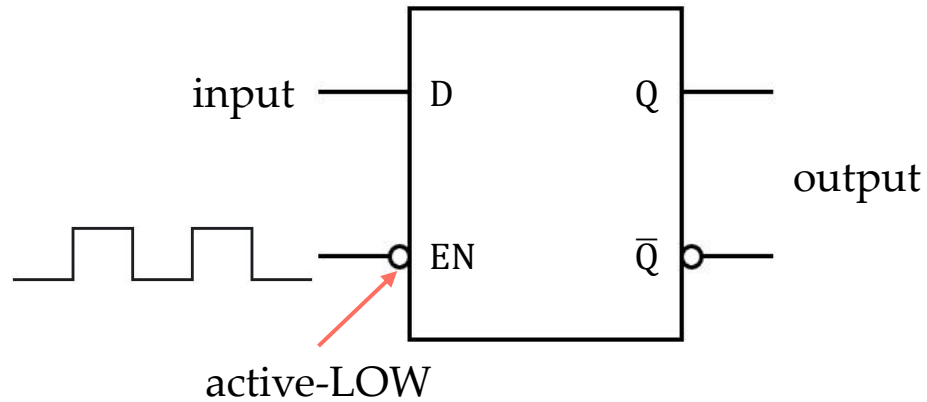


Flip-flop is an edge-triggered (rising or falling) device.

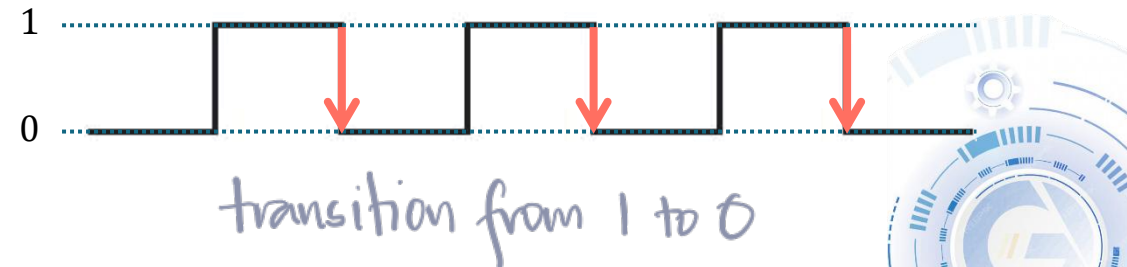
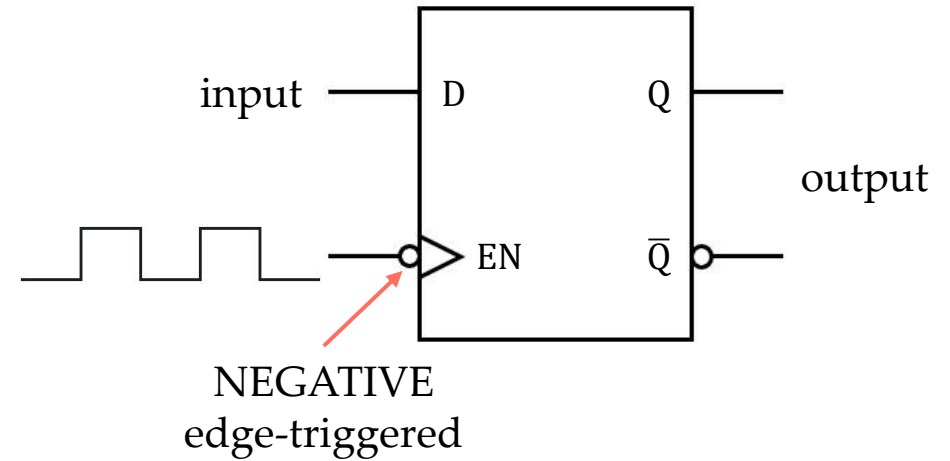


LATCH AND FLIP-FLOP

Latch is a level sensitive (HIGH or LOW) device.

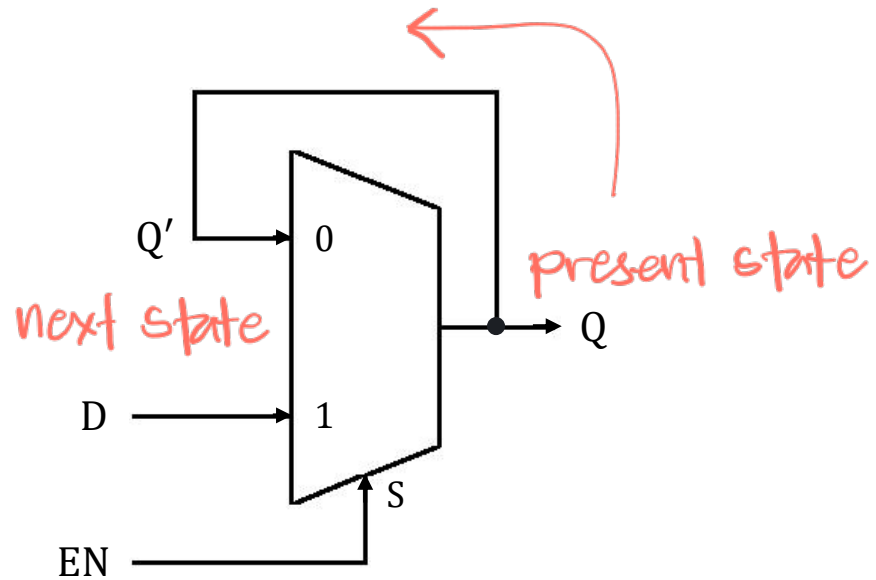


Flip-flop is an edge-triggered (rising or falling) device.



BASIC LATCH

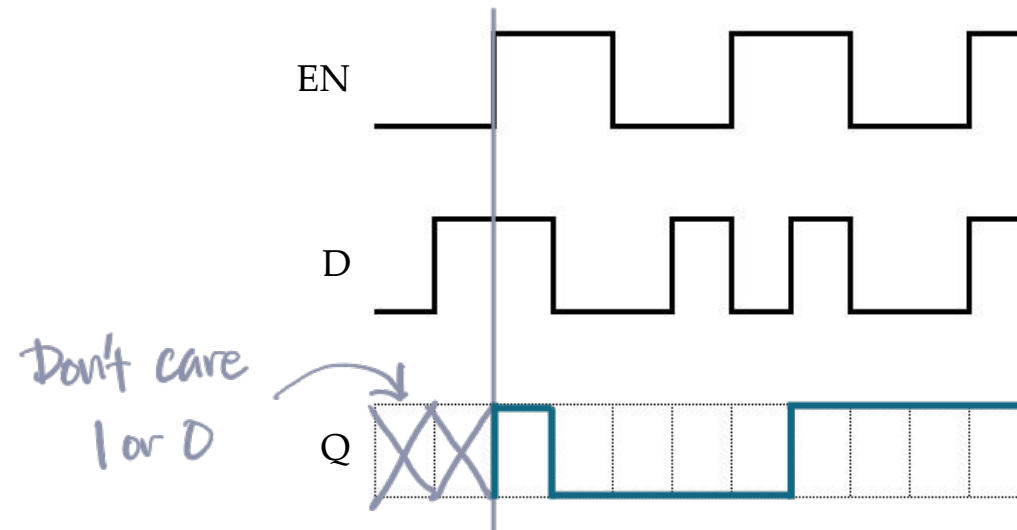
2-to-1 MUX as settable storage element



Characteristic Table

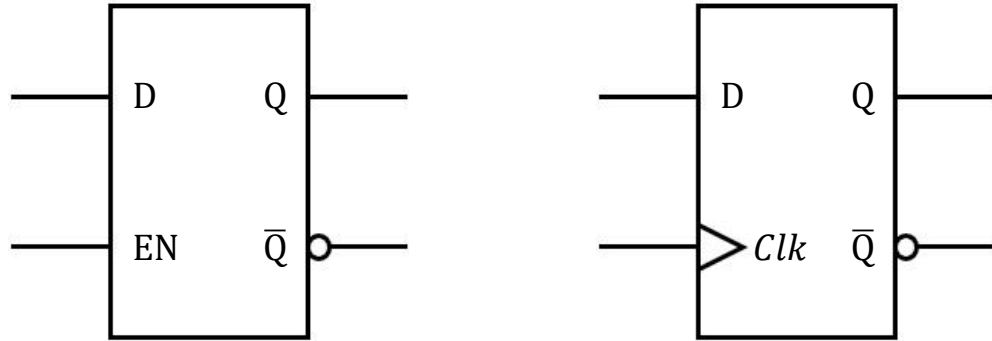
EN	D	Q
0	X	Q'
1	0	0
1	1	1

Timing Diagram



D LATCH/FLIP-FLOP

Graphical Symbol



Characteristic Table

EN	D	Q
0	X	NC
1	0	0
1	1	1

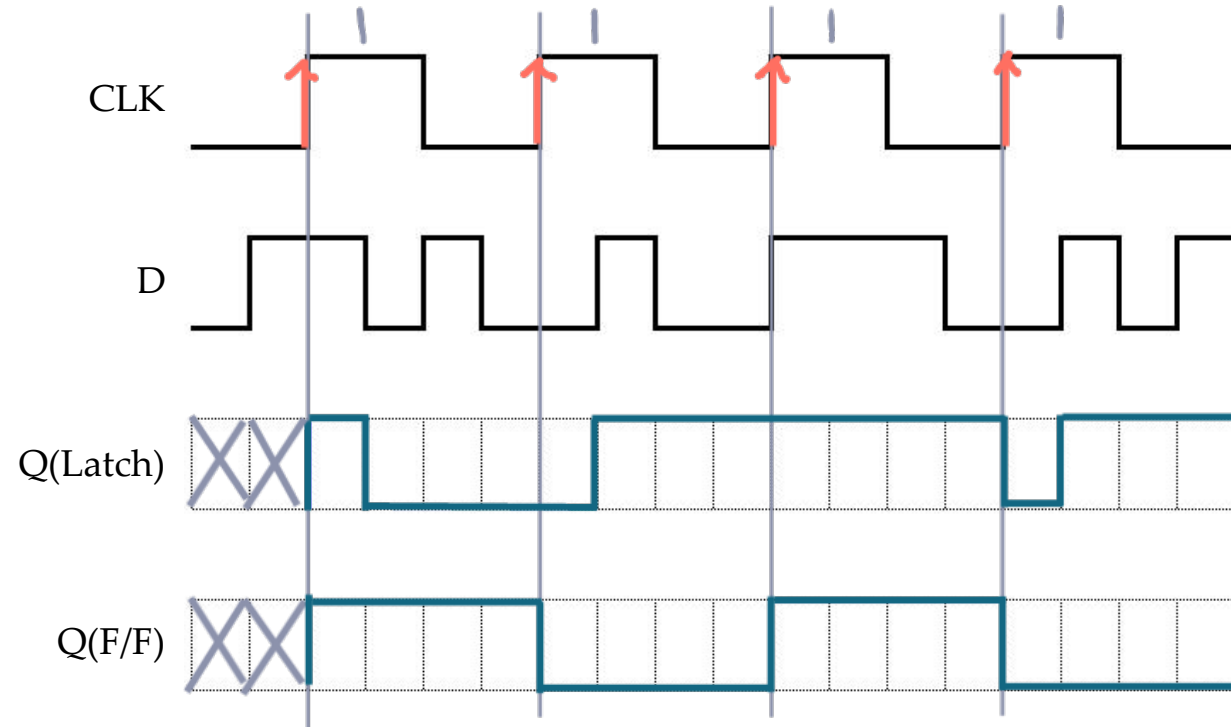
← latch

Clk	D	Q(t+1)
0/1	X	Q(t)
↑	0	0
↑	1	1

Q(t) – present state

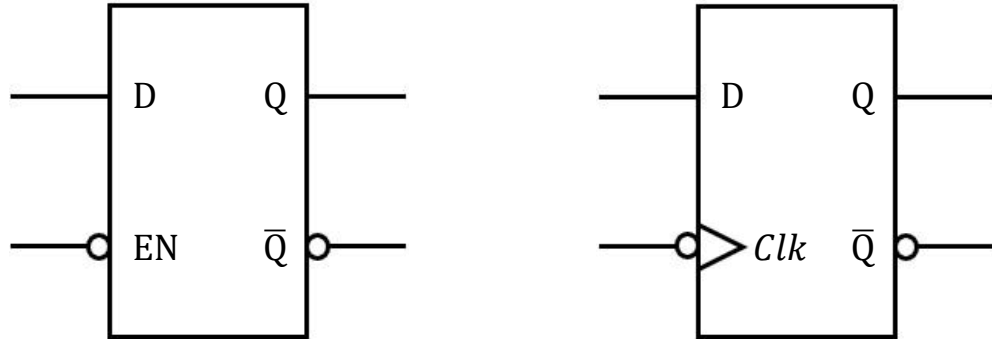
Q(t+1) – next state

Timing Diagram



D LATCH/FLIP-FLOP

Graphical Symbol



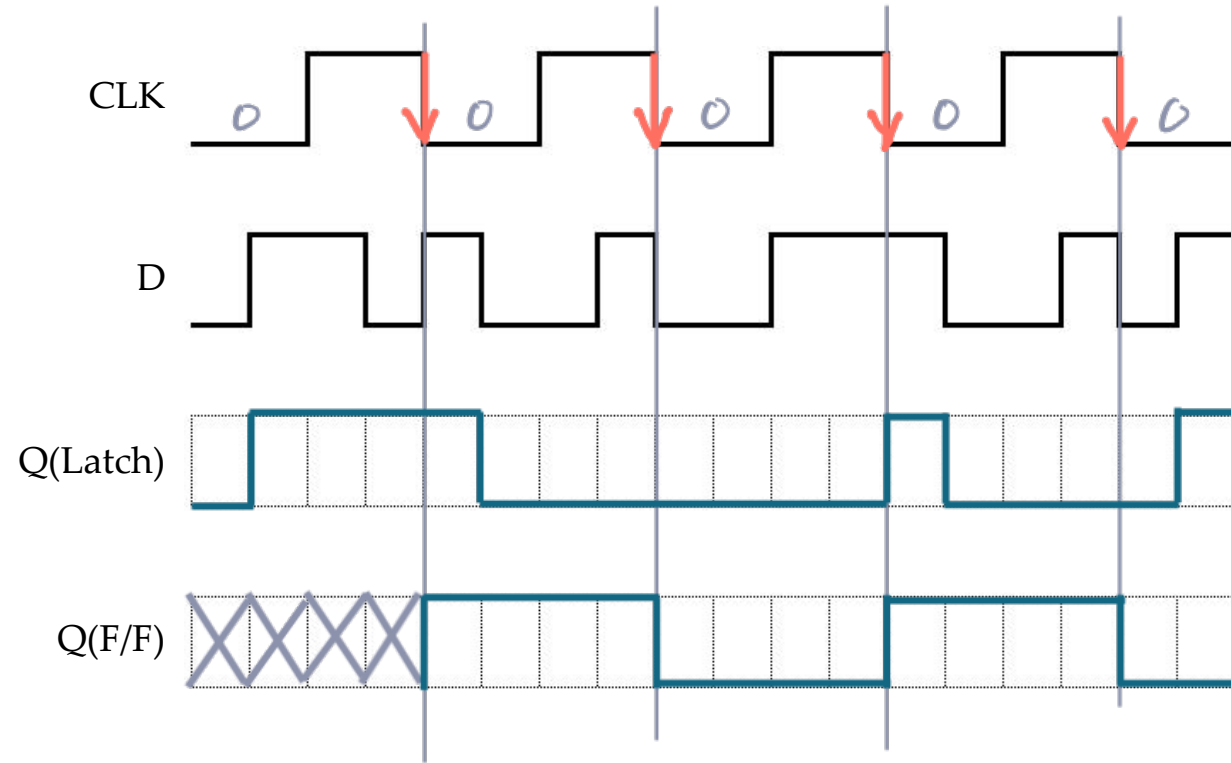
Characteristic Table

EN	D	Q
1	X	NC
0	0	0
0	1	1

Clk	D	Q(t+1)
0/1	X	Q(t)
↓	0	0
↓	1	1

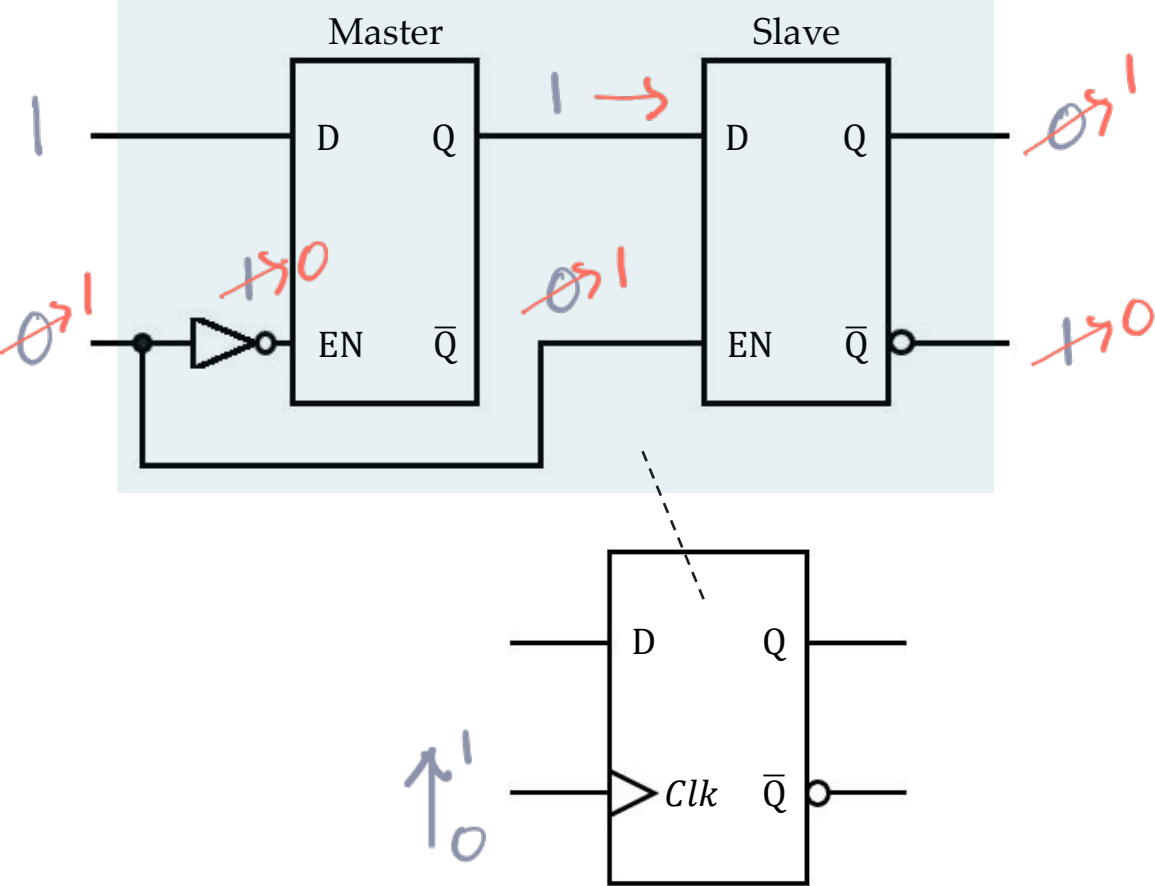
Q(t) – present state
Q(t+1) – next state

Timing Diagram

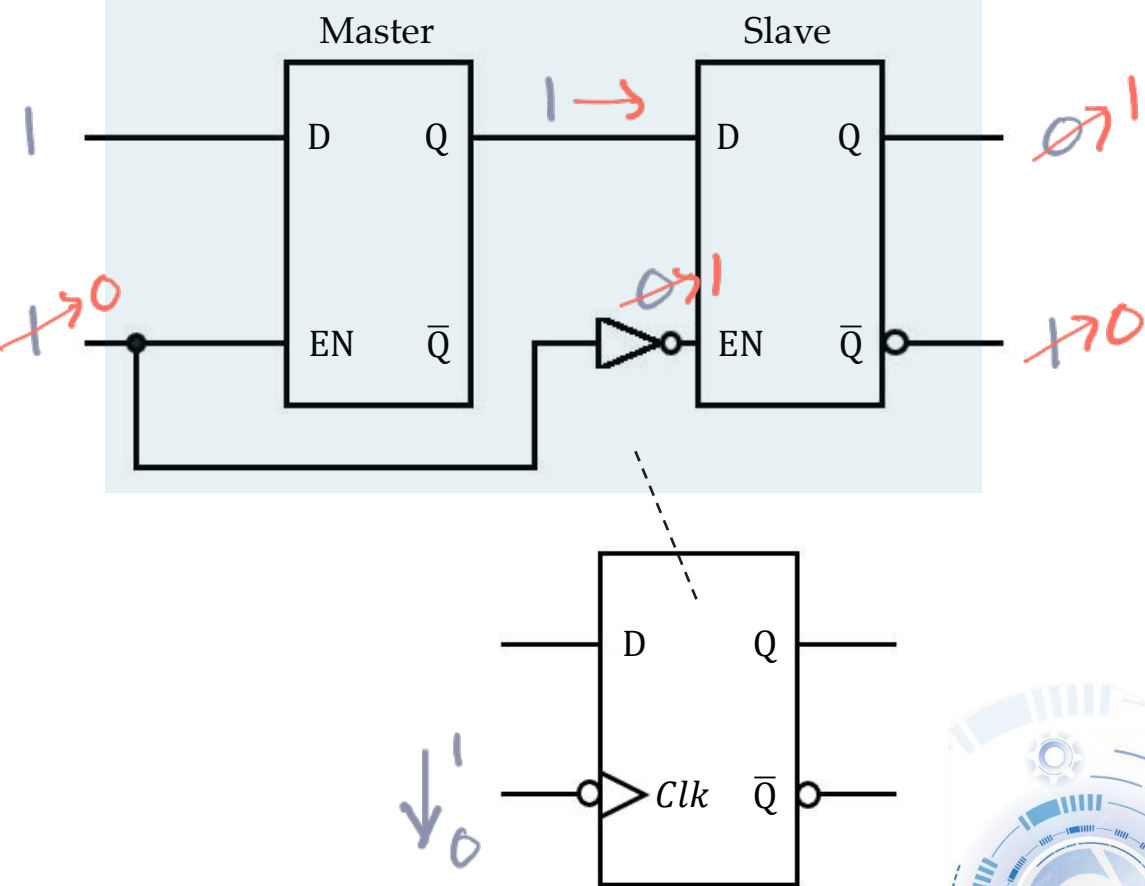


MASTER-SLAVE D FLIP-FLOP

Positive edge-triggered

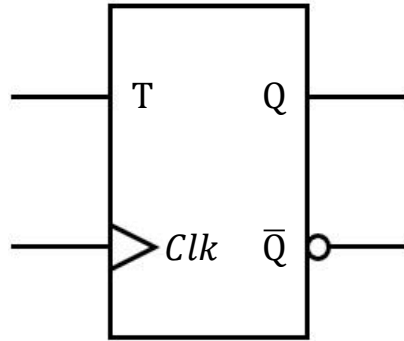


Negative edge-triggered



T FLIP-FLOP

Graphical Symbol

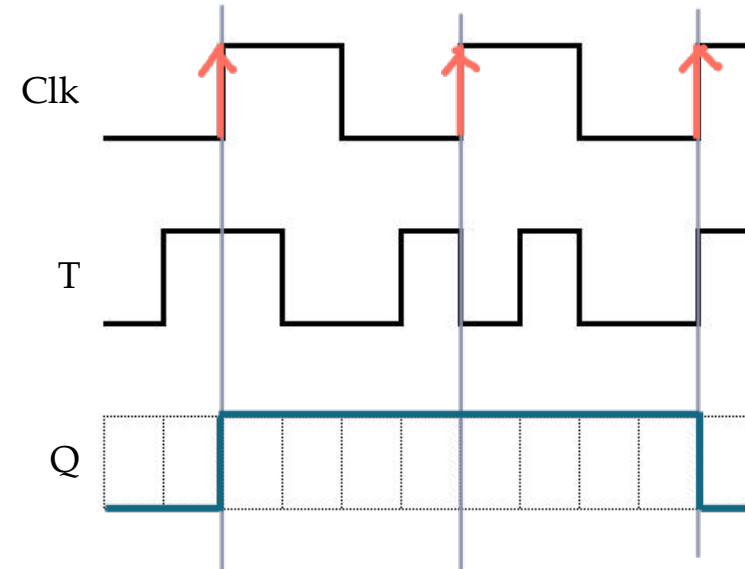


Characteristic Table

T	$Q(t+1)$
0	$Q(t)$
1	$\bar{Q}(t)$

← toggles

Timing Diagram

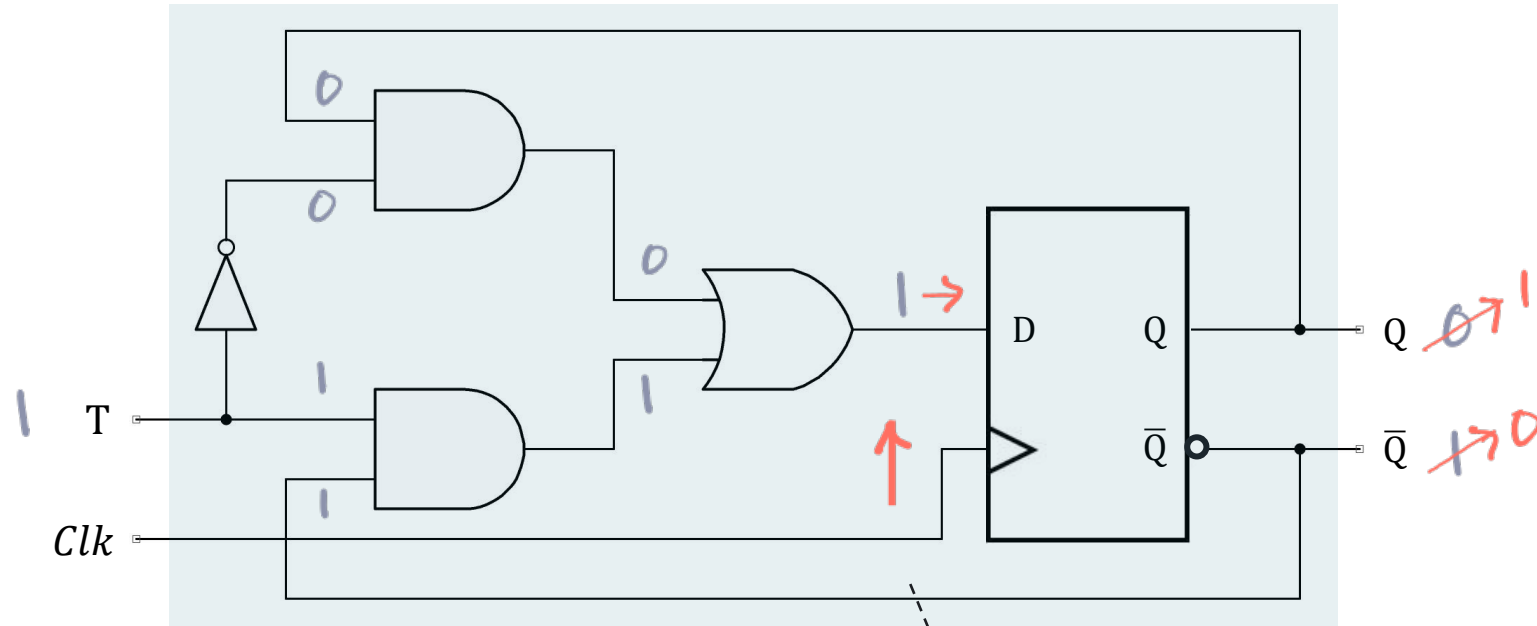


Let initial state = 0

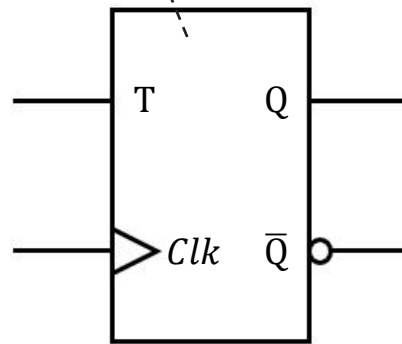


T FLIP-FLOP

Implementation



When $T=1$, output toggles

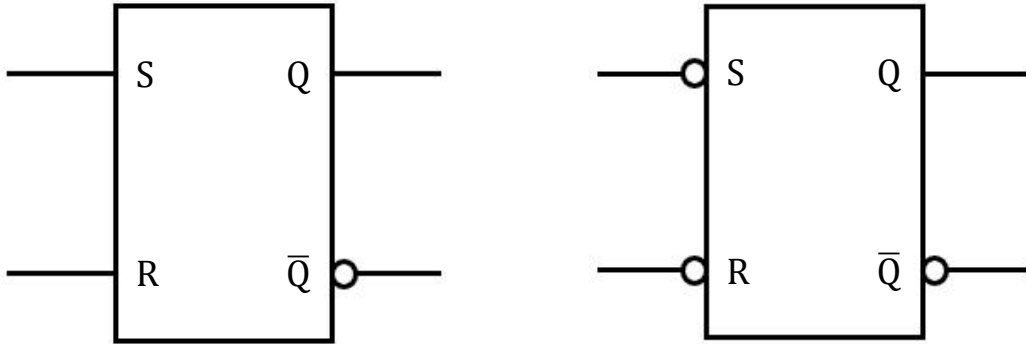


SR LATCH



SR LATCH

Graphical Symbol



Set $\rightarrow Q = 1$

Reset $\rightarrow Q = 0$

Characteristic Table

S	R	Q	$\bar{Q}$	Remark
0	0	NC	NC	No Change
0	1	0	1	Reset
1	0	1	0	Set
1	1	0	0	Invalid

Active-HIGH

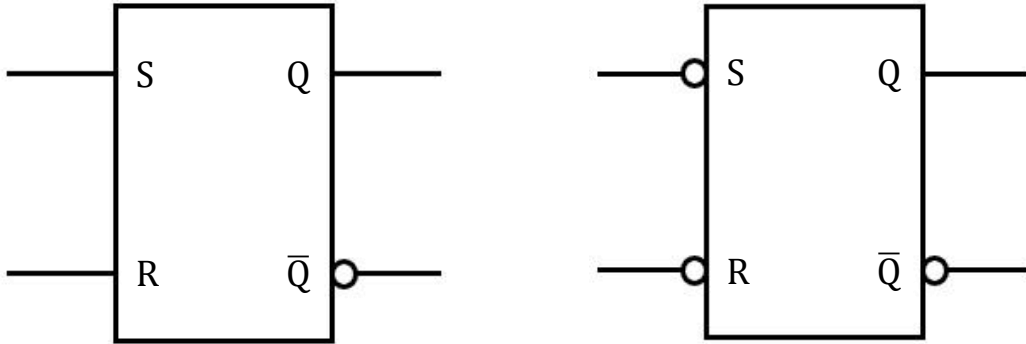
S	R	Q	$\bar{Q}$	Remark
0	0	1	1	Invalid
0	1	1	0	Set
1	0	0	1	Reset
1	1	NC	NC	No Change

Active-LOW



SR LATCH

Graphical Symbol

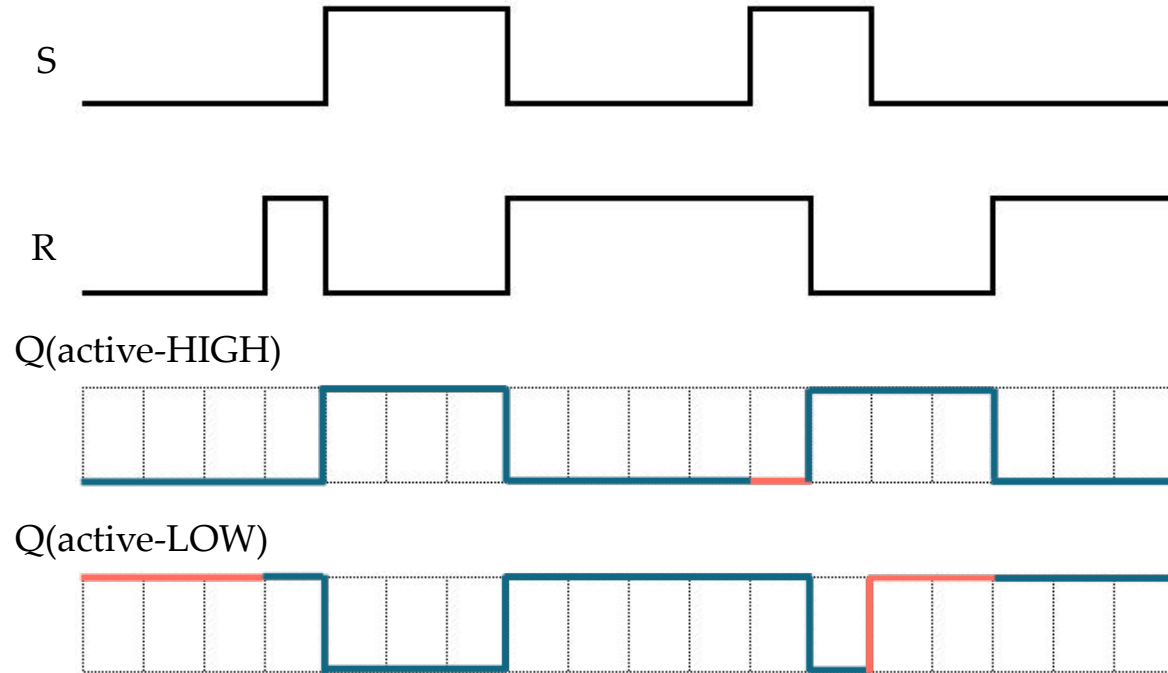


Characteristic Table

S	R	Q	$\bar{Q}$	Remark
0	0	NC	NC	No Change
0	1	0	1	Reset
1	0	1	0	Set
1	1	0	0	Invalid

Active-HIGH

Timing Diagram

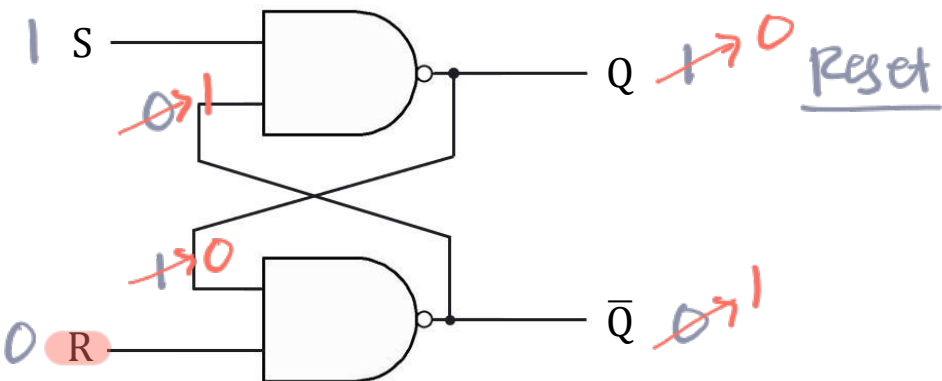
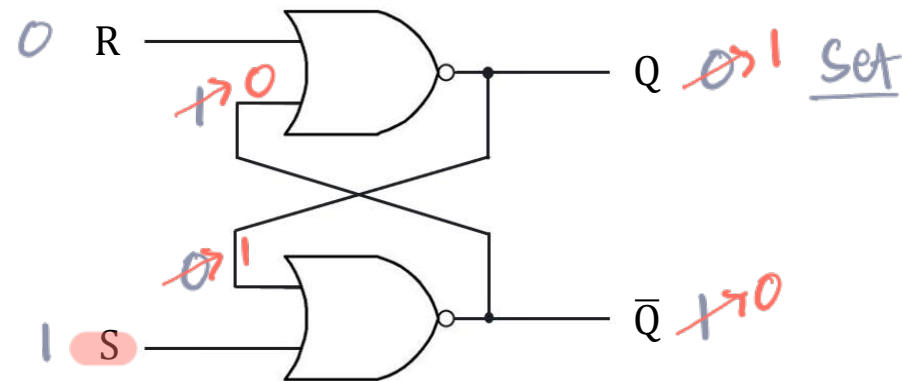


Let initial state = 0



SR LATCH

Implementation



Characteristic Table

S	R	Q	$\bar{Q}$	Remark
0	0	NC	NC	No Change
0	1	0	1	Reset
1	0	1	0	Set
1	1	0	0	Invalid

Active-HIGH

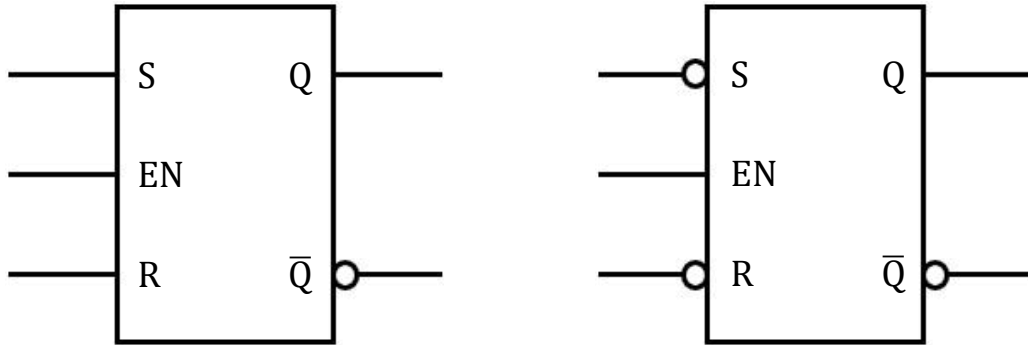
S	R	Q	$\bar{Q}$	Remark
0	0	1	1	Invalid
0	1	1	0	Set
1	0	0	1	Reset
1	1	NC	NC	No Change

Active-LOW



GATED SR LATCH

Graphical Symbol



Both are active-HIGH enable

Characteristic Table

EN	S	R	Q	$\bar{Q}$	Remark
0	X	X	Q	$\bar{Q}$	No Change
1	0	0	Q	$\bar{Q}$	No Change
1	0	1	0	1	Reset
1	1	0	1	0	Set
1	1	1	0	0	Invalid

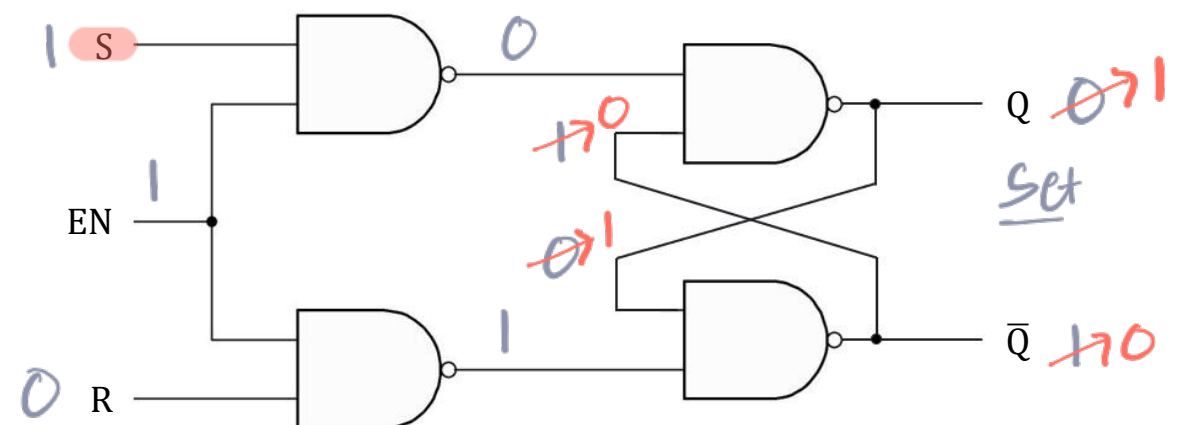
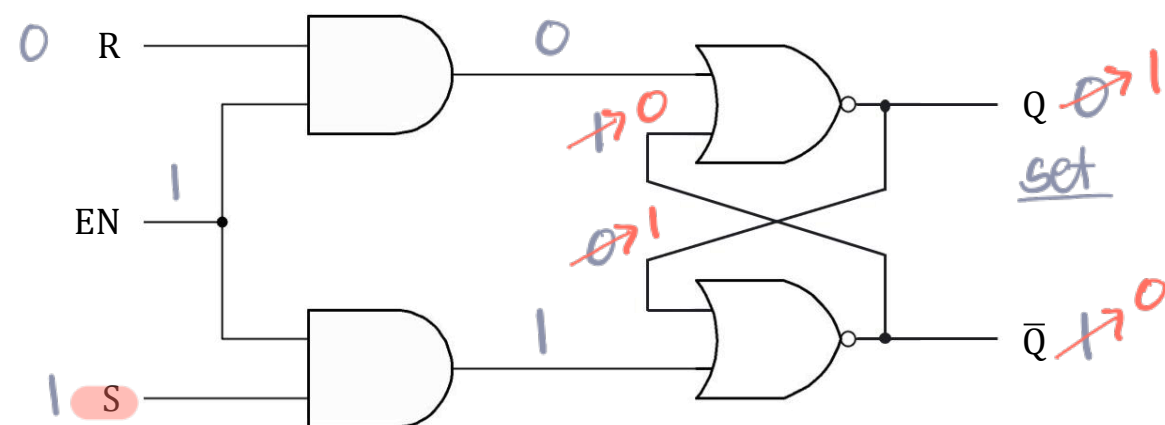
Active-HIGH

EN	S	R	Q	$\bar{Q}$	Remark
0	X	X	Q	$\bar{Q}$	No Change
1	0	0	1	1	Invalid
1	0	1	1	0	Set
1	1	0	0	1	Reset
1	1	1	Q	$\bar{Q}$	No Change

Active-LOW

GATED SR LATCH

Implementation



Characteristic Table

EN	S	R	Q	$\bar{Q}$	Remark
0	X	X	Q	$\bar{Q}$	No Change
1	0	0	Q	$\bar{Q}$	No Change
1	0	1	0	1	Reset
1	1	0	1	0	Set
1	1	1	0	0	Invalid

Active-HIGH

EN	S	R	Q	$\bar{Q}$	Remark
0	X	X	Q	$\bar{Q}$	No Change
1	0	0	1	1	Invalid
1	0	1	1	0	Set
1	1	0	0	1	Reset
1	1	1	Q	$\bar{Q}$	No Change

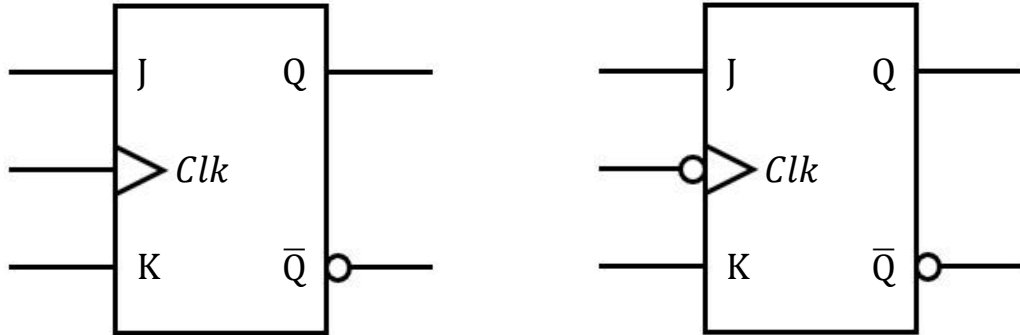
Active-LOW

JK FLIP-FLOP



JK FLIP-FLOP

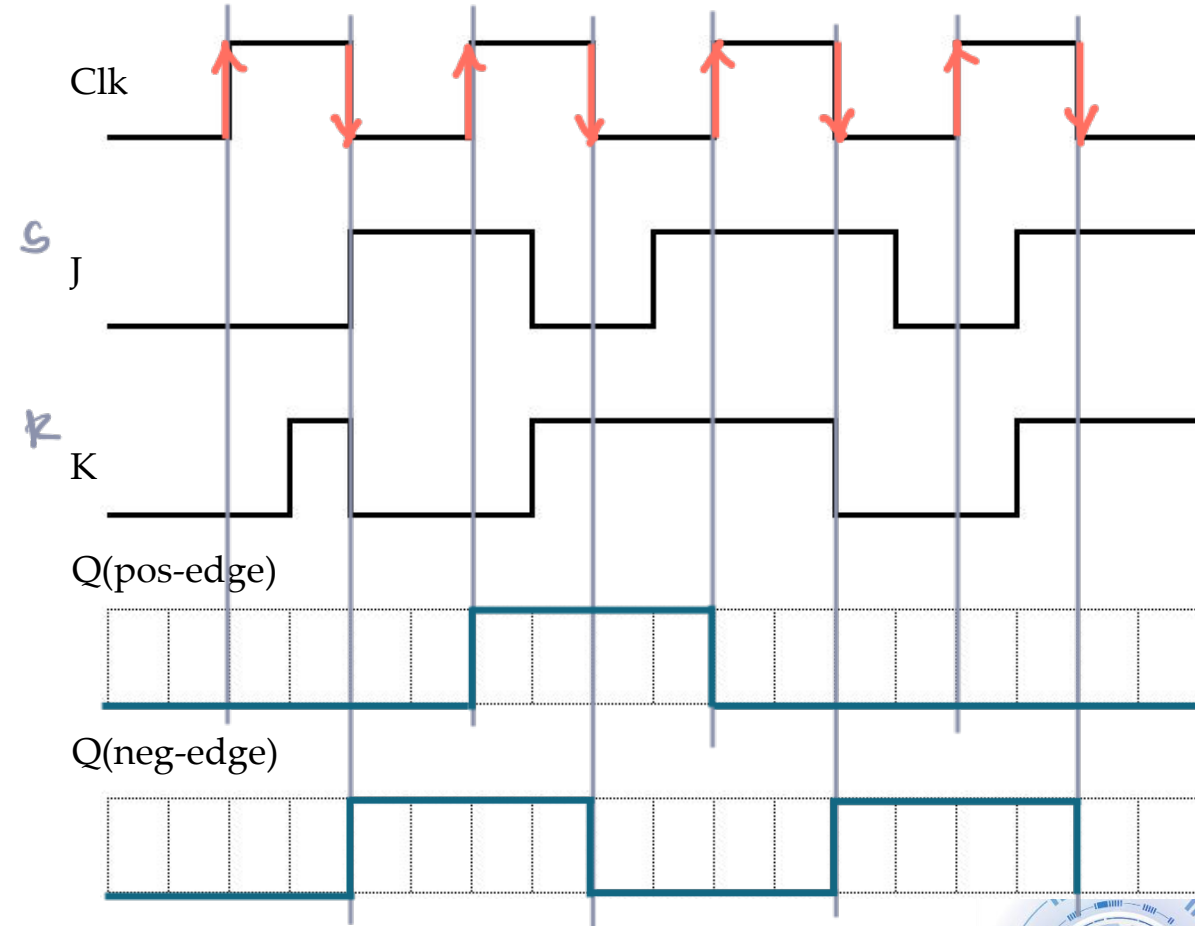
Graphical Symbol



Characteristic Table

J	K	$Q(t+1)$	Remark
0	0	$Q(t)$	No Change
0	1	0	Reset
1	0	1	Set
1	1	$\bar{Q}(t)$	Toggle

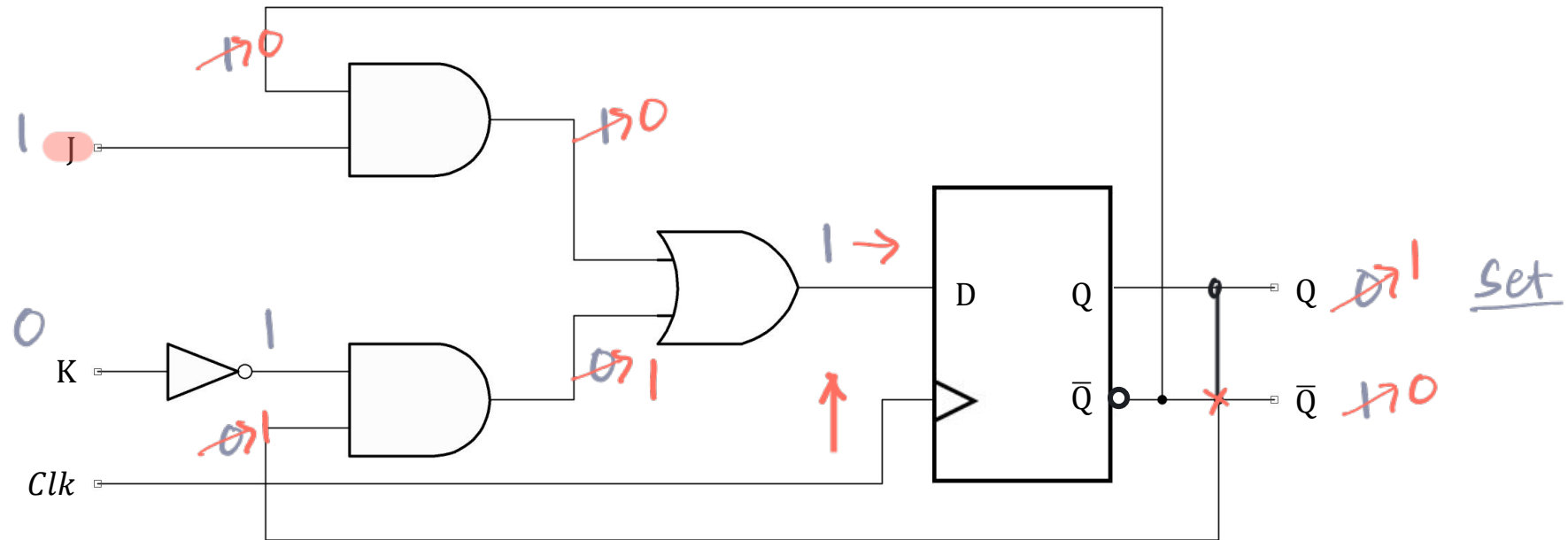
Timing Diagram



let initial state = 0

JK FLIP-FLOP

Implementation



LABORATORY

